DASC 3240, L08

Barplot and more – Assignments

2023/24 SPRING SEMESTER

MASAYUKI USHIO, ASSISTANT PROFESSOR AT OCES
USHIO@UST.HK (RM CYT-2013)

Today's outline

- Barplot and more – Assignments
  - Data
  - Assignments

Data

Datasets

Rdatasets 1.0.0 Available datasets

What is this?

`Rdatasets` is a collection of 2264 datasets which were originally distributed alongside the statistical software environment `R` and some of its add-on packages. The goal is to make these data more broadly accessible for teaching and statistical software development.

What is included?

The list of available datasets (csv and docs) is available here:

- HTML index
- CSV index

On the github repository you will also find the scripts I use to scrape data and update the website.

https://vincentarelbundock.github.io/Rdatasets/

ChickWeight

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/ChickWeight.html
https://vincentarelbundock.github.io/Rdatasets/doc/datasets/ChickWeight.csv

Weight versus age of chicks on different diets

Description

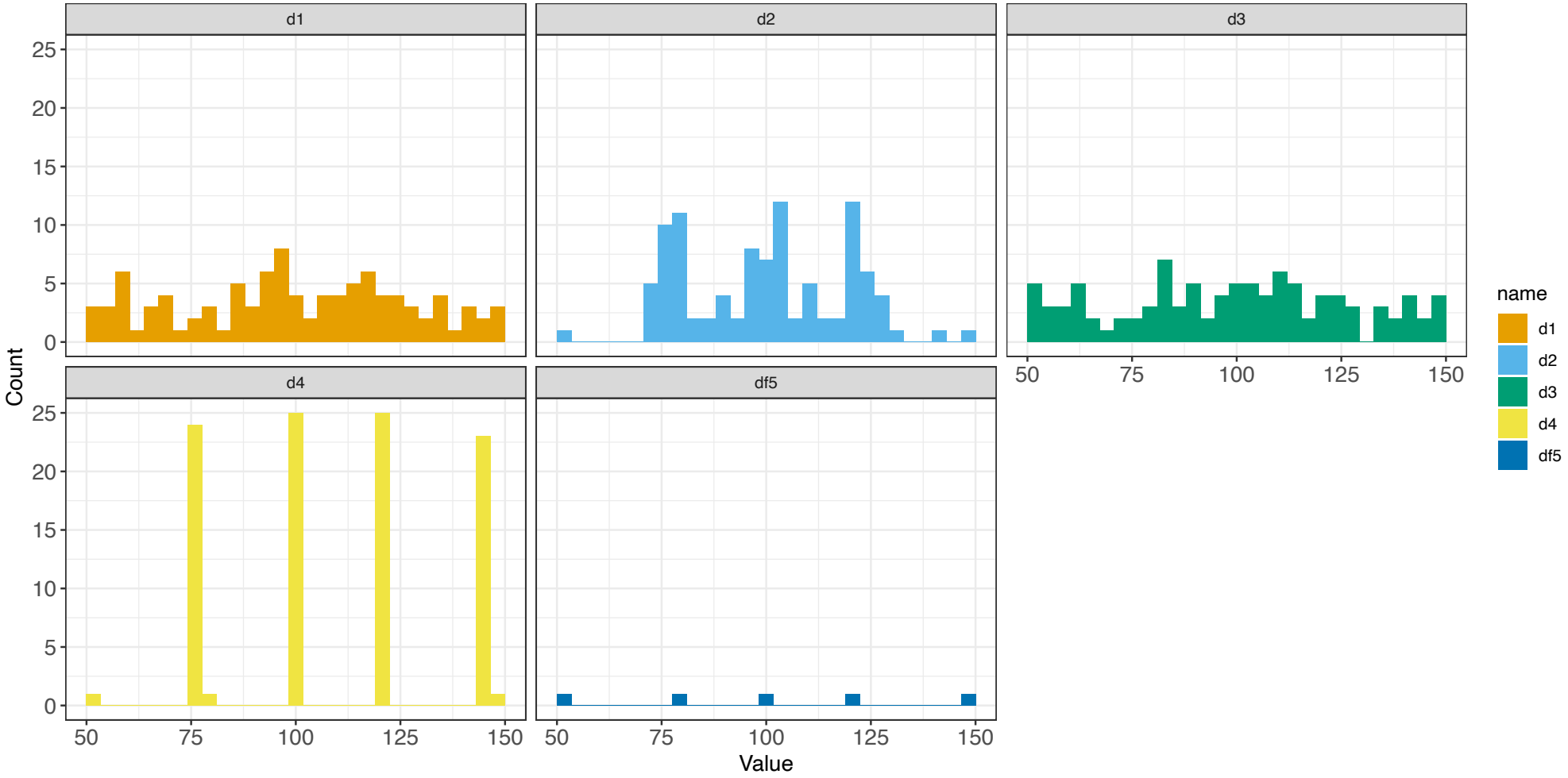
The ChickWeight data frame has 578 rows and 4 columns from an experiment on the effect of diet on early growth of chicks.

	weight	Time	Chick	Diet
1	42	0	1	1
2	51	2	1	1
3	59	4	1	1
4	64	6	1	1
5	76	8	1	1
6	93	10	1	1

weight the body weight of the chick (gm).
Time the number of days since birth when the measurement was made.
Chick levels 18 < ... < 48 giving a unique identifier for the chick. The ordering of the levels groups chicks on the same diet together and orders them according to their final weight (lightest to heaviest) within diet.
Diet levels 1, ..., 4 indicating which experimental diet the chick received.

Custom datasets

The custom datasets created by myself: d1, d2, d3, d4, and d5



https://github.com/ong8181/DASC3240/blob/main/data/d1.csv ~ d5.csv

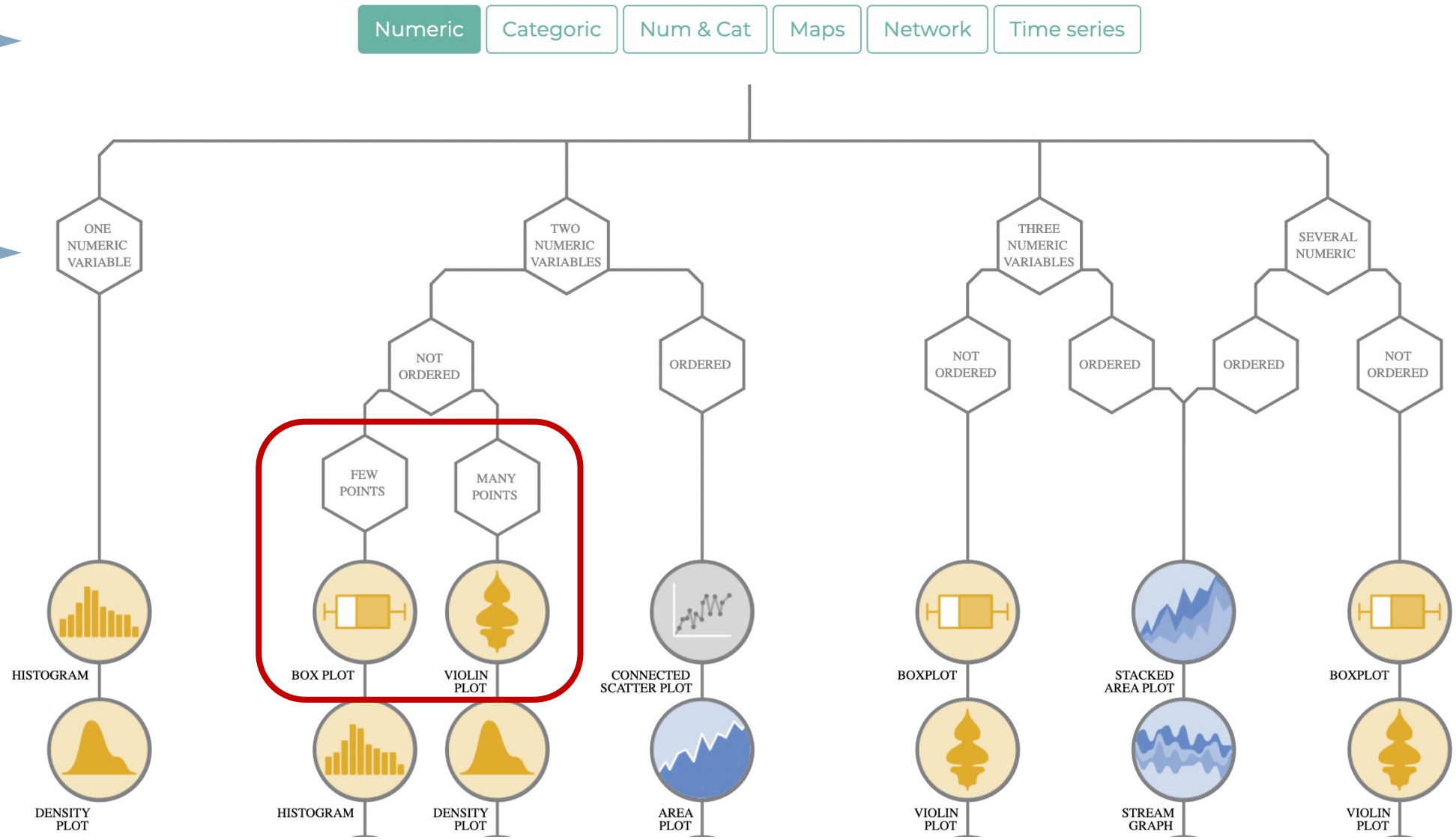
Assignments

BARPLOTS AND MORE

How many variables? Numeric or categorical?

Numeric or categorical?

The number of variables



https://www.data-to-viz.com/

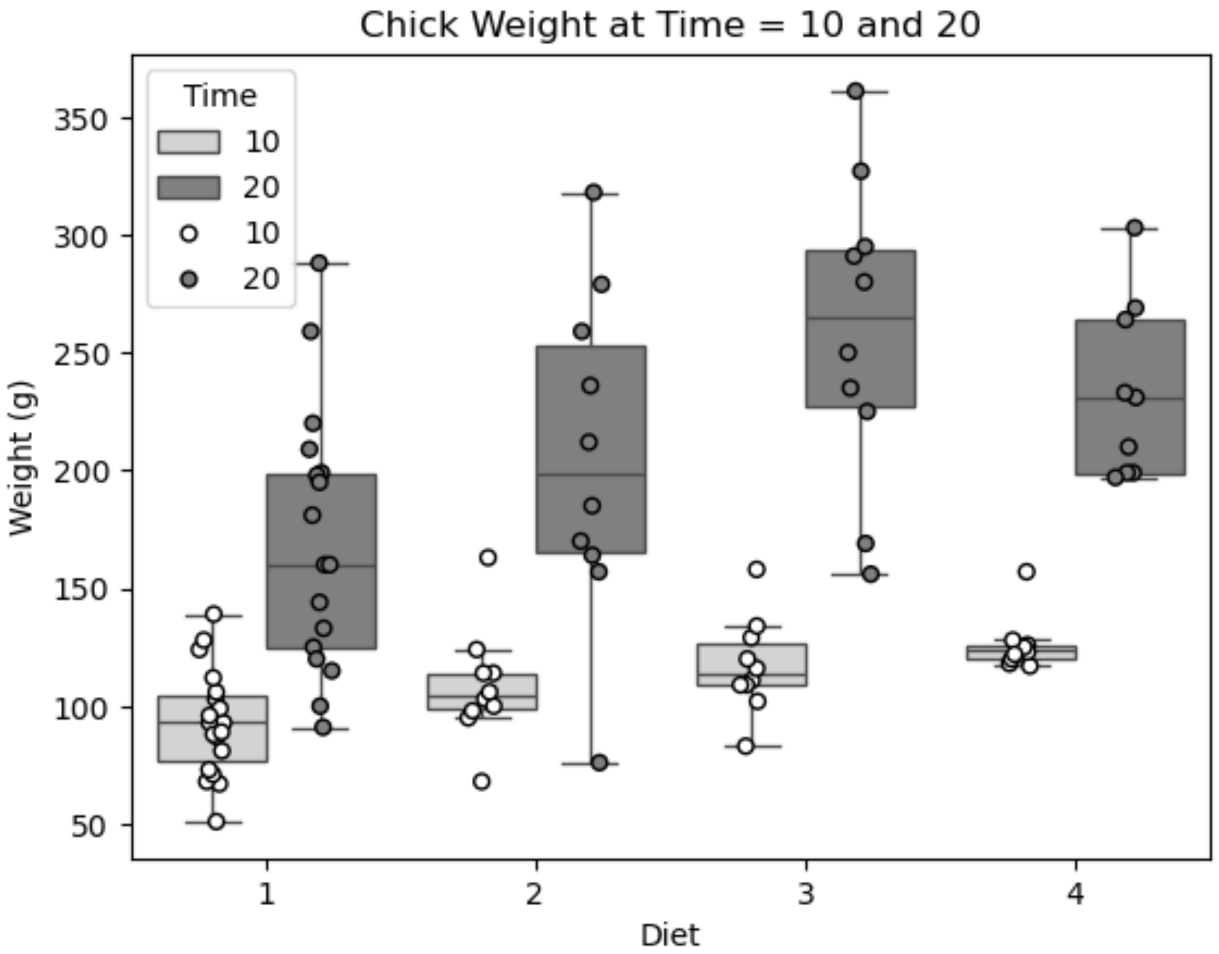
Assignments

- **ChickWeight**
  1. Extract data with **Time = 10** and visualize the effect of **Diet** on **weight** using boxplots overlaid with jitter plots (please be careful of outlier points).
  2. Extract data with **Time = 20** and visualize the effect of **Diet** on **weight** using boxplots overlaid with jitter plots (please be careful of outlier points).
  3. Combine the two figures generated in **Task 1** and **Task 2** so that readers can easily understand the effects of **Diet** and **Time** on **weight** (an example is provided).
- **Custom datasets**
  1. Combine the custom datasets (**d1**, **d2**, **d3**, **d4**, and **d5**) into one 'pandas.DataFrame'. Visualize them using boxplots overlaied with beeswarm plots (an example is provided).

Assignments

- **ChickWeight**

  3. Combine the two figures generated in **Task 1** and **Task 2** so that readers can easily understand the effects of **Diet** and **Time** on **weight** (an example is provided).

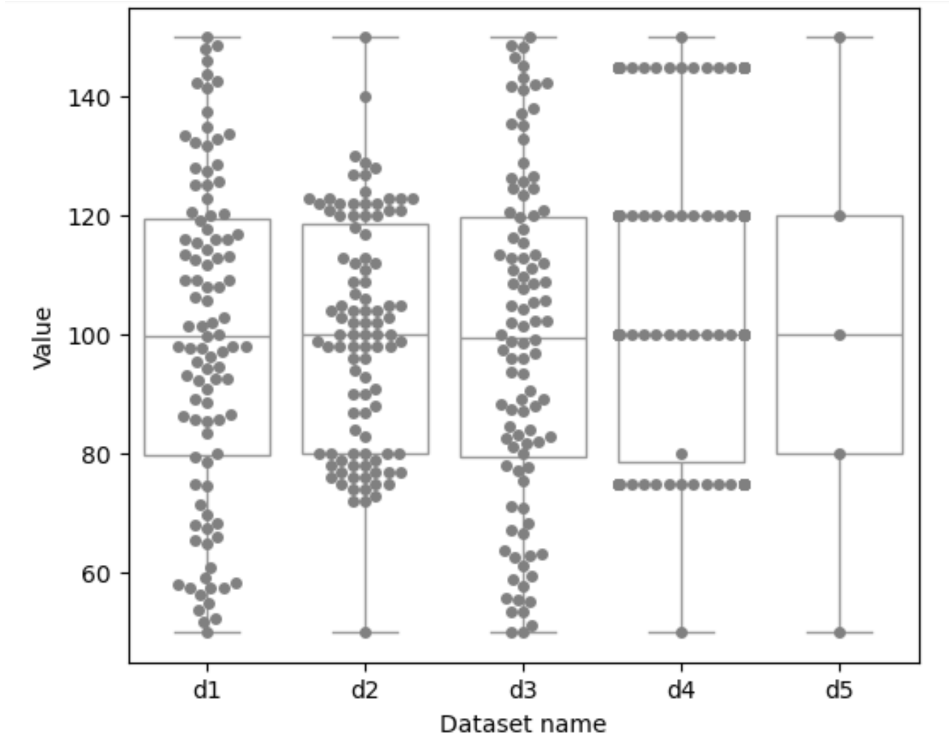


Assignments

- **Custom datasets**
  1. Combine the custom datasets into one ‘pandas.DataFrame’. Visualize them using boxplots overlaied with beeswarm plots (an example is provided).

Tidy-data

	id	value	dataset
0	1	150.000000	d1
1	2	125.302795	d1
2	3	147.974703	d1
3	4	122.979472	d1
4	5	128.737786	d1
...	...	...	...
0	1	150.000000	d5
1	2	120.000000	d5
2	3	100.000000	d5
3	4	80.000000	d5
4	5	50.000000	d5



https://cran.r-project.org/web/packages/tidyr/vignettes/tidy-data.html

Tidy data

- **Tidy data**
  - "It is often said that **80%** of data analysis is spent on the cleaning and preparing data"
  - "Tidy datasets provide a standardized way to **link the structure of a dataset (its physical layout) with its semantics (its meaning)**."

In tidy data:

1. Each variable is a column; each column is a variable.
2. Each observation is a row; each row is an observation.
3. Each value is a cell; each cell is a single value.

https://cran.r-project.org/web/packages/tidyr/vignettes/tidy-data.html

Tidy data

Messy data:

`relig_income` contains religion, income and frequency

Religion	<$10k	$10–20k	$20–30k	$30–40k	...	>150k
Buddhist	27	21	30	43	...	53
Catholic	418	617	732	670	...	633
...	...	...	...	...	...	...
Unaffiliated	211	299	374	365	...	258

This is a variable

Tidy data:

A variable should be in a column

Religion	income	frequency
Buddhist	<$10k	27
Buddhist	$10–20k	21
Buddhist	$20–30k	30
...	...	...

Tidy data

Mexico weather dataset

Day of the month

Names of variables

	A	B	C	D	E	F	G	H	I		
1	id	year	month	element	d1	d2	d3	d4	d5	d6	d7
2	MX17004	2010	1	TMAX	NA	NA	NA	NA	NA	NA	NA
3	MX17004	2010	1	TMIN	NA	NA	NA	NA	NA	NA	NA
4	MX17004	2010	2	TMAX	NA	273	241	NA	NA	NA	NA
5	MX17004	2010	2	TMIN	NA	144	144	NA	NA	NA	NA
6	MX17004	2010	3	TMAX	NA	NA	NA	NA	321	NA	NA
7	MX17004	2010	3	TMIN	NA	NA	NA	NA	142	NA	NA
8	MX17004	2010	4	TMAX	NA	NA	NA	NA	NA	NA	NA
9	MX170		4	TMIN	NA	NA	NA	NA	NA	NA	NA
10	MX17004	2010	5	TMAX	NA	NA	NA	NA	NA	NA	NA
11	MX17004	2010	5	TMIN	NA	NA	NA	NA	NA	NA	NA
12	MX17004	2010	6	TMAX	NA	NA	NA	NA	NA	NA	NA
13	MX17004	2010	6	TMIN	NA	NA	NA	NA	NA	NA	NA
14	MX17004	2010	7	TMAX	NA	NA	286	NA	NA	NA	NA
15	MX17004	2010	7	TMIN	NA	NA	175	NA	NA	NA	NA
16	MX17004	2010	8	TMAX	NA	NA	NA	NA	296	NA	NA
17	MX17004	2010	8	TMIN	NA	NA	NA	NA	158	NA	NA
18	MX17004	2010	10	TMAX	NA	NA	NA	NA	270	NA	
19	MX17004	2010	10	TMIN	NA	NA	NA	NA	140	NA	

Tidy data

Mexico weather dataset

Names of variables

Mostly tidy

id	year	month	day	element	value
MX17004	2010	1	30	tmax	27.8
MX17004	2010	1	30	tmin	14.5
MX17004	2010	2	2	tmax	27.3
MX17004	2010	2	3	tmax	24.1
MX17004	2010	2	11	tmax	29.7
MX17004	2010	2	23	tmax	29.9

Tidy

id	year	month	day	tmax	tmin
MX17004	2010	1	30	27.8	14.5
MX17004	2010	2	2	27.3	14.4
MX17004	2010	2	3	24.1	14.4
MX17004	2010	2	11	29.7	13.4
MX17004	2010	2	23	29.9	10.7
MX17004	2010	3	5	32.1	14.2

1. Each variable is a column
2. Each observation is a row
3. Each value is a cell

Tidy data

- **In Excel sheet (or CSV) (if you want to use it for data analysis)**
  - NEVER insert line feeds within a cell
  - NEVER add more than one value in a cell
  - NEVER merge cells
  - Using only alphabets is often a safe strategy

Tidy data

Religion	<$10k	$10–20k	$20–30k	$30–40k	...	>150k
Buddhist	27	21	30	43	...	53
Catholic	418	617	732	670	...	633
...	...	...	...	...	...	...
Unaffiliated	211	299	374	365	...	258

- Sometimes, untidy format is convenient for data input (by human)
- But you still need to think about how you will make the data tidy
- **If the original data is already tidy, mostly tidy, or easy to be tidied, it will save 80% of data analysis time, which makes your (data analysis) life happy!**